STUDENT ENGINEERING PROJECT

SMART ACCESS CONTROL SYSTEM

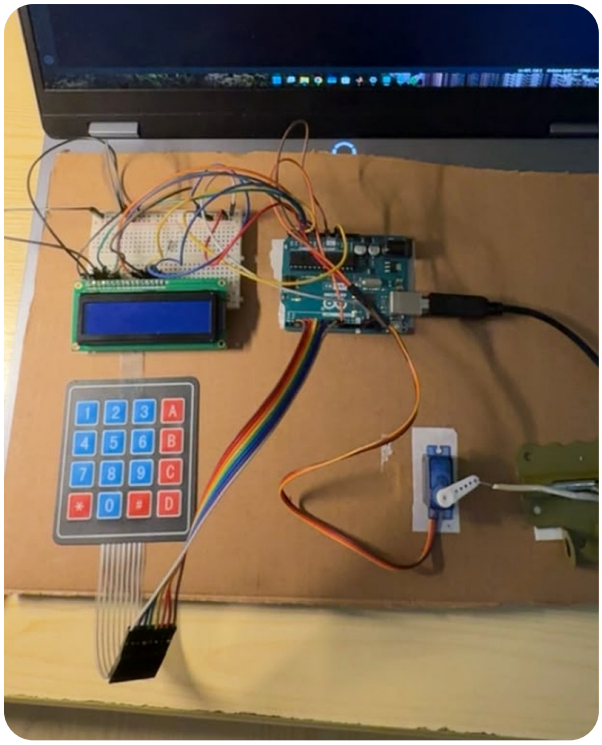
Arduino-based PIN authentication with a servo-actuated barrier

PROJECT OVERVIEW

A physical, educational access-control prototype that accepts a four-digit PIN through a matrix keypad and controls a servo-operated locking mechanism. An LCD and buzzer provide status feedback throughout the interaction.

Core engineering areas

- Embedded C++ and event-driven input
- Hardware / software integration
- Nonvolatile configuration with EEPROM
- User-interface and actuator control



INPUT
4 x 4 keypad

DECISION
PIN verification

OUTPUT
Servo + LCD + buzzer

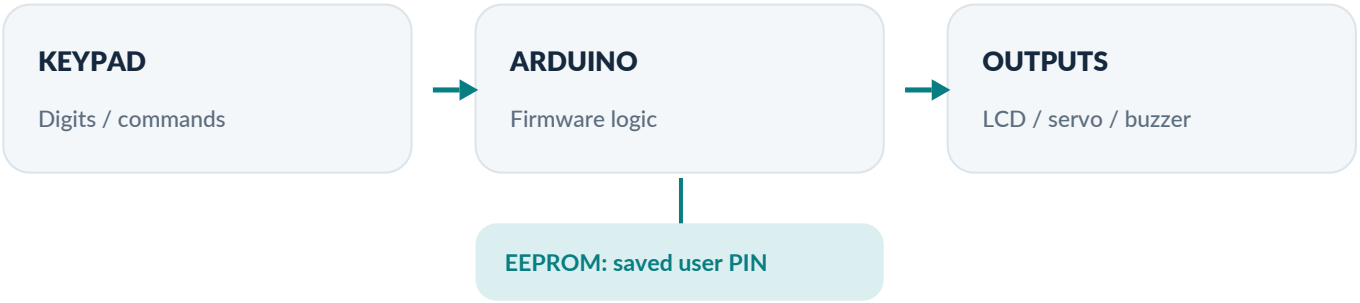
PROJECT CONTEXT

Academic hardware prototype. This document describes the supplied source code and filmed prototype; it does not certify the system as a secure or production-ready access-control device.

HARDWARE & ARCHITECTURE

A compact microcontroller system with explicit user input, control decisions, feedback, and mechanical actuation.

FUNCTIONAL BLOCK DIAGRAM



PIN ASSIGNMENTS IN THE SOURCE

Keypad rows	D9, D8, D7, D6
Keypad columns	D5, D4, D3, D2
LCD 16x2 (4-bit)	RS A0 | E A1 | D4-D7 A2-A5
Servo signal	D10
Buzzer	D11

USER INTERACTION FLOW

01
ENTER
User types a 4-digit code

02
VERIFY
Firmware checks the PIN

03
ACTUATE
Servo opens barrier

04
RESET
Barrier closes; idle resumes

INTERFACE NOTES

Digits 0-9 enter the PIN; C deletes a digit; # confirms; * opens the administrator flow after its code is entered; D requests PIN-protected service mode. This mapping is read from the Arduino source.

FIRMWARE, TESTING & NEXT STEPS

Functional behaviors below are identified in the supplied source. Full build validation and hardware regression testing remain pending.

IMPLEMENTED LOGIC

PIN & ADMINISTRATION

Four-digit entry, confirmation, administrator flow and PIN update.

PERSISTENT STORAGE

EEPROM initialization, validation and saving of the user PIN.

ACCESS PROTECTION

Three unsuccessful entries lead to a 10-second temporary lockout.

TIMEOUT HANDLING

Partial PIN entry expires after 10 seconds of inactivity.

SERVO BEHAVIOR

Progressive 0° to 90° motion; an open-state countdown precedes closure.

SERVICE OPERATION

PIN-protected manual operation with a 15-second timeout.

EVIDENCE & VERIFICATION

AVAILABLE ARTIFACTS

Arduino source code, photograph of the assembled prototype and a short demonstration video.

NOT YET VERIFIED

Clean compilation, full function-by-function hardware testing, power-cycle persistence and edge cases.

ENGINEERING LIMITATIONS / ROADMAP

01 **Blocking execution**

Several functions use delay() or busy loops; a non-blocking state machine would improve responsiveness.

02 **Memory predictability**

Dynamic String use can be replaced by fixed-size character buffers on memory-constrained boards.

03 **Security and safety**

Demo credentials and plaintext EEPROM are not suitable for real access protection; additional fail-safe design is required.